

4.5.3 Forces and elasticity

Content	Key opportunities for skills development
Students should be able to: • give examples of the forces involved in stretching, bending or compressing an object • explain why, to change the shape of an object (by stretching, bending or compressing), more than one force has to be applied – this is limited to stationary objects only • describe the difference between elastic deformation and inelastic deformation caused by stretching forces. The extension of an elastic object, such as a spring, is directly proportional to the force applied, provided that the limit of proportionality is not exceeded.	

Content	Key opportunities for skills development
$force = spring\ constant \times extension$ $[F = k e]$ force, F , in newtons, N spring constant, k , in newtons per metre, N/m extension, e , in metres, m	MS 3b, c, 4a Students should be able to recall and apply this equation.
This relationship also applies to the compression of an elastic object, where 'e' would be the compression of the object. A force that stretches (or compresses) a spring does work and elastic potential energy is stored in the spring. Provided the spring is not inelastically deformed, the work done on the spring and the elastic potential energy stored are equal. Students should be able to: - describe the difference between a linear and non-linear relationship between force and extension - calculate a spring constant in linear cases	MS 3b, c, 4a
interpret data from an investigation of the relationship between force and extension	WS 3.5
calculate work done in stretching (or compressing) a spring (up to the limit of proportionality) using the equation: $elastic\ potential\ energy = 0.5 \times spring\ constant \times (extension)^2$ $[E_e = \frac{1}{2} k e^2]$	MS 3c Students should be able to apply this equation which is given on the Physics equation sheet. This equation is also given in Changes in energy (page 18).
Students should be able to calculate relevant values of stored energy and energy transfers.	MS 3c

Required practical activity 6: investigate the relationship between force and extension for a spring.

AT skills covered by this practical activity: AT 1 and 2.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 95).